

5. Continuous Random Variables

Learning Objectives

- By the end of this week, you should be able to:

- Understand *continuous RVs* & how they differ from discrete RVs
- Calculate probabilities of *continuous* RVs using *Probability Density Functions* (*PDF*)
- Calculate *expectations* of continuous RVs
- Employ common continuous distributions:
  - *Uniform, Exponential, Gamma, Normal*

Continuous RVs

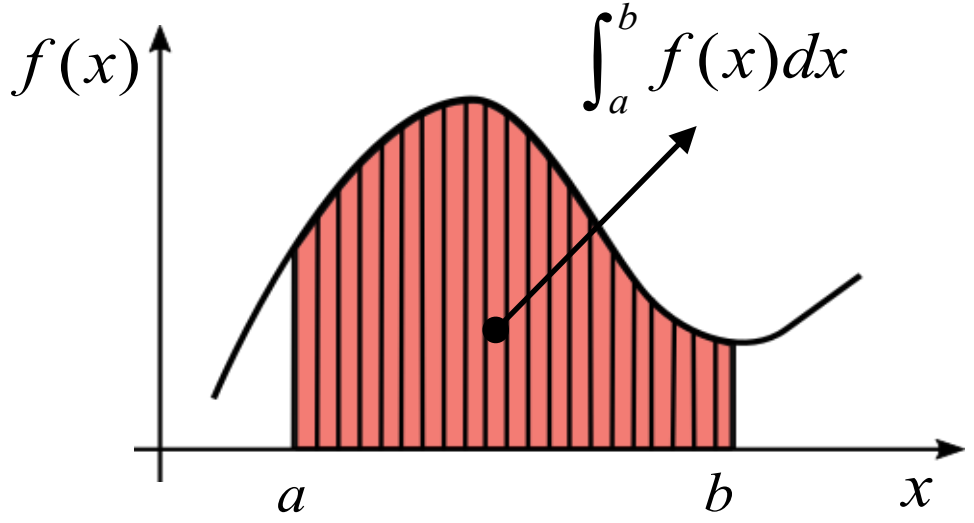
- RV X called *continuous* if $P(X=x)=0$, $\forall\, x\in\mathbb{R}$ (i.e. PMF=0)
  - Can $P(S)=1$, when $P(X=x)=0$, $\forall\, x\in\mathbb{R}$?
  - *YES*, if X takes *uncountably* many values.
    - E.g. can have $P(X\in[a,b])>0$ for real interval $[a,b]$.

- Continuous RVs assume *uncountable* number of values
  - Cannot create list $\{x_1,x_2,...\}$ of possible values
  - E.g. X is waiting time at bus stop $\rightarrow X\in\mathbb{R}^+$

- Calculate probabilities of continuous RVs using Calculus

Continuous RVs

- How do you measure area in Euclidean space?
- Think of area as *uncountable* collection of *lines*
  - Each line has area 0
    - Same holds for *countably* many lines
  - But *uncountably* many lines can have area >0 !



- Calculus: area is given by definite integral $\int_a^b f(x)dx$
  - Function $f(x)$ represents line *height*

Probabilities for Continuous RVs

- For continuous RV X, mainly interested in probabilities of *intervals* of values of X

$$P(X \in (a, b]) = P(a < X \leq b), \forall a < b \in \mathbb{R}$$

  - Interval type (closed/open) does not matter

$$P(X \in (a, b]) = P(X \in (a, b)) = P(X \in [a, b)) = P(X \in [a, b])$$

- Calculate interval probabilities in two ways
  - Cumulative Distribution Functions (CDF)
  - *Probability Density Functions (PDF)*

Probability Density Function

- For continuous RV X, its *Probability Density Function* (*PDF*) is function $f_X(\cdot)$ such that

$$P(X \in [a,b]) = \int_a^b f_X(x)dx$$

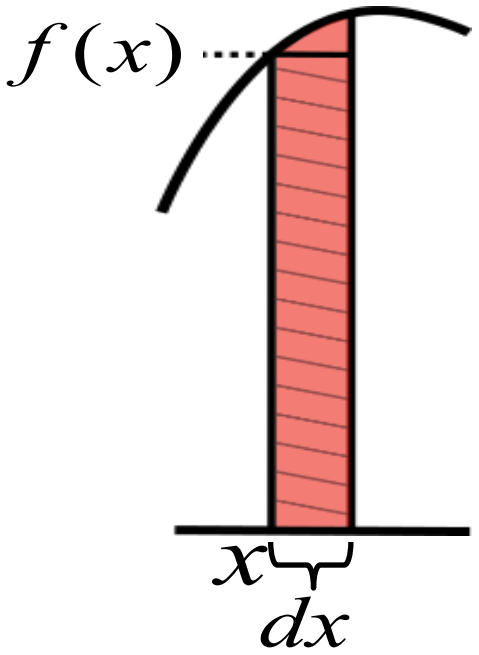
- Properties of PDFs
  - $0 \leq f_X(x), \forall x \in \mathbb{R}$ (can have $f(x) > 1$!)
  - $\int_{-\infty}^{+\infty} f_X(x)dx = 1 \quad (= P(-\infty < X < +\infty) = P(S))$
  - $F_X(x) = \int_{-\infty}^{x} f_X(u)du \Rightarrow f_X(x) = \frac{d}{dx}F_X(x) = F'_X(x)$ (PDF $\leftrightarrow$ CDF)

Probability Density Function

- PDF gives rate at which probability accumulates around value x of RV X

$$P(x < X \leq x + dx) = F(x + dx) - F(x) \approx$$
$$\approx F'(x)dx = f(x)dx, \quad \text{for } dx \approx 0$$

- Higher density $f(x)$ means higher probability for intervals around x



Example

- Consider RV X with PDF $f(x)=\begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
  - Find the probability $P(2<X<4)$
  - Find the corresponding CDF

Uniform Distribution

- *Uniform* RV X takes values in interval $[l,u]$, $l<u\in\mathbb{R}$, so that probability of any subinterval (a,b) is *proportional to its length*

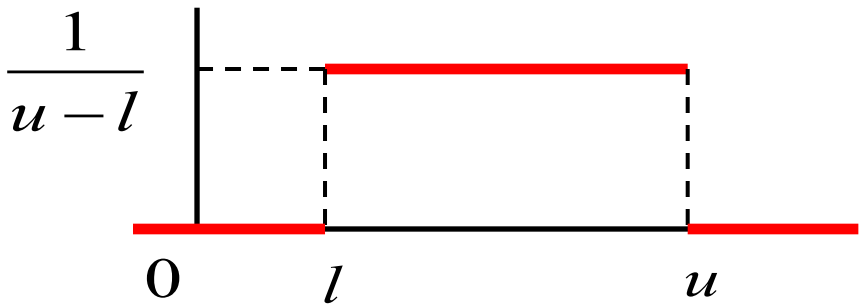
$$P(a<X<b)=\frac{b-a}{u-l},\quad \forall l\le a\le b\le u$$

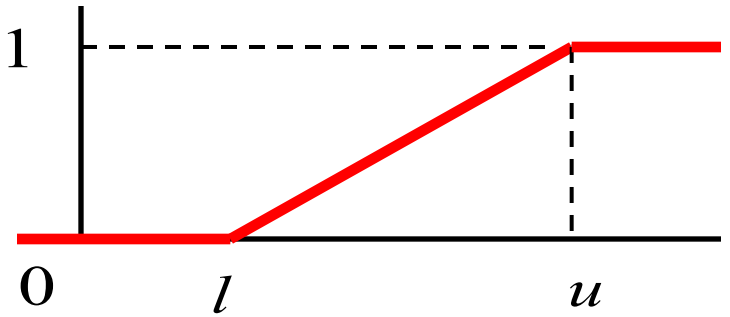
  - Denoted X~Uniform(l,u)
  - PDF

$$f(x)=\begin{cases}\dfrac{1}{u-l}, & l\le x\le u\\ 0, & \text{otherwise}\end{cases}$$

  - CDF

$$F(x)=\begin{cases}0, & x<l\\ \dfrac{x-l}{u-l}, & l\le x\le u\\ 1, & x>u\end{cases}$$





Example

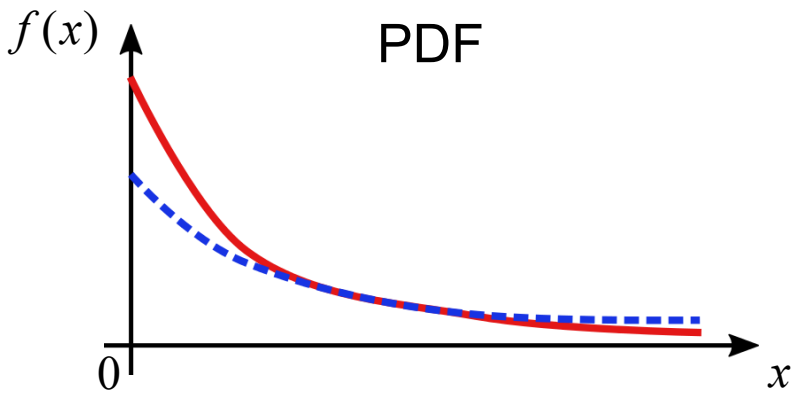
- Bus passes every 30min and you arrive at bus stop at random time. Let RV X be your waiting time.
  - What is the distribution of X?
  - What is the probability you wait more than 20min?
  - What is the probability you wait more than 20min, given that you have already waited for 10min?

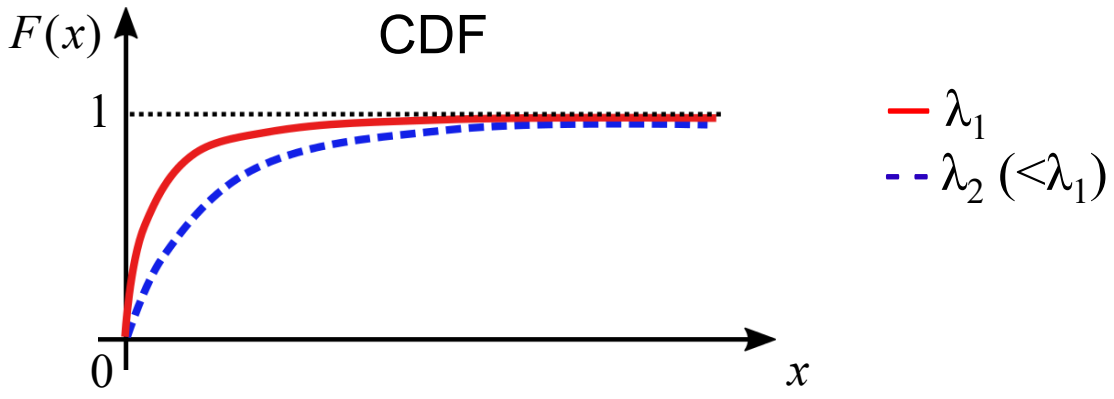
Exponential Distribution

- *Exponential* RV X takes positive values according to PDF

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \text{for some } \lambda > 0$$

  - Typically measures *time* until something happens
  - Denoted by X~Exponential(λ)





Example

- Service time X of request to web server follows Exponential(λ)
  - Find the CDF of the service time
  - Show that $P(X > x + y \mid X > y) = P(X > x)$
    (i.e. Exponential is *memory-less*)

Expected Values - Continuous RVs

- For *continuous* RV X with PDF $f_X(x)$, expected values given by

$$\boxed{E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx}$$

(if integral converges)

  - Mean given by $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$

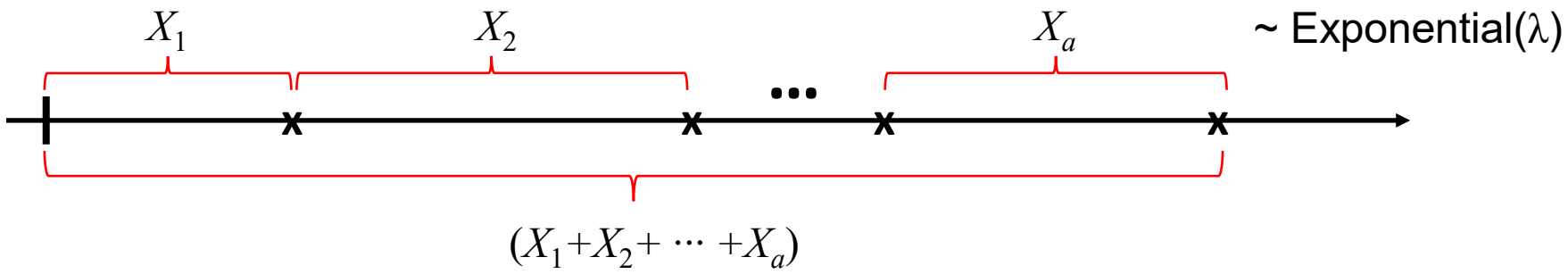
- All properties of expectations hold in continuous case
  - E.g., linearity $E(aX + bY) = aE(X) + bE(Y)$

Example

- Find the expected value of the Exponential(λ) distribution

Gamma Distribution

- Sequence of #a *independent* Exponential(λ) values



- Sum of independent exponentials follows *Gamma* distribution
  - Used to model *positive* quantities: e.g., system lifetime, total insurance claims, aggregate rainfall, etc.
  - (Continuous analogue of Negative Binomial)

Gamma Distribution

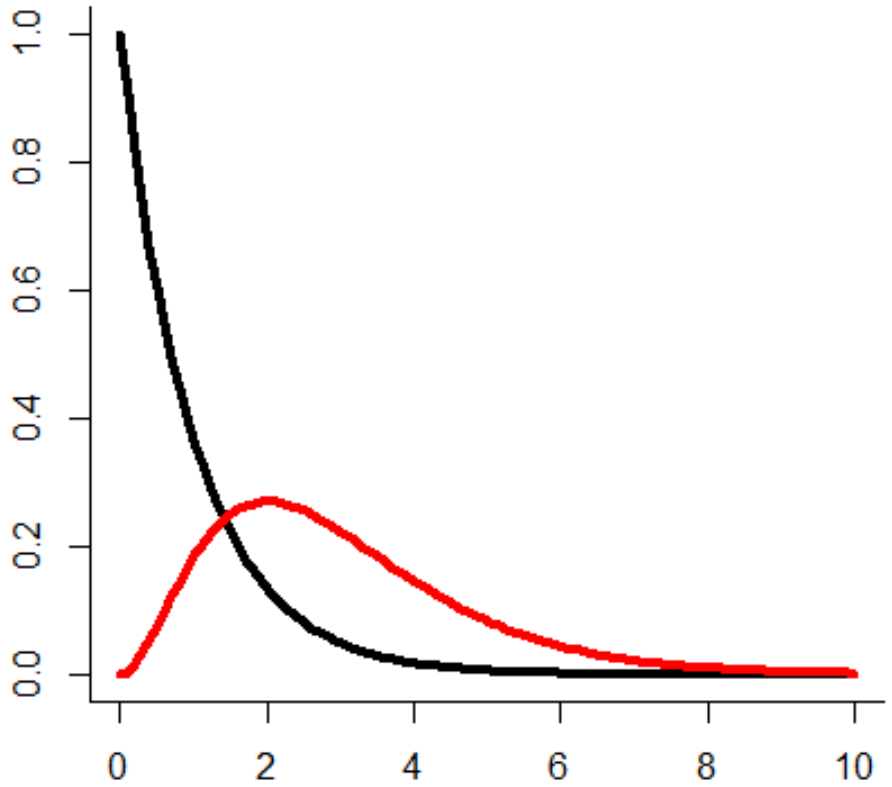
- PDF of Gamma distribution is
  - Parameters $a,\lambda > 0$, where $a,\lambda \in \mathbb{R}$
  - Denoted $X \sim \text{Gamma}(a,\lambda)$

$$f(x) = \begin{cases} \frac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

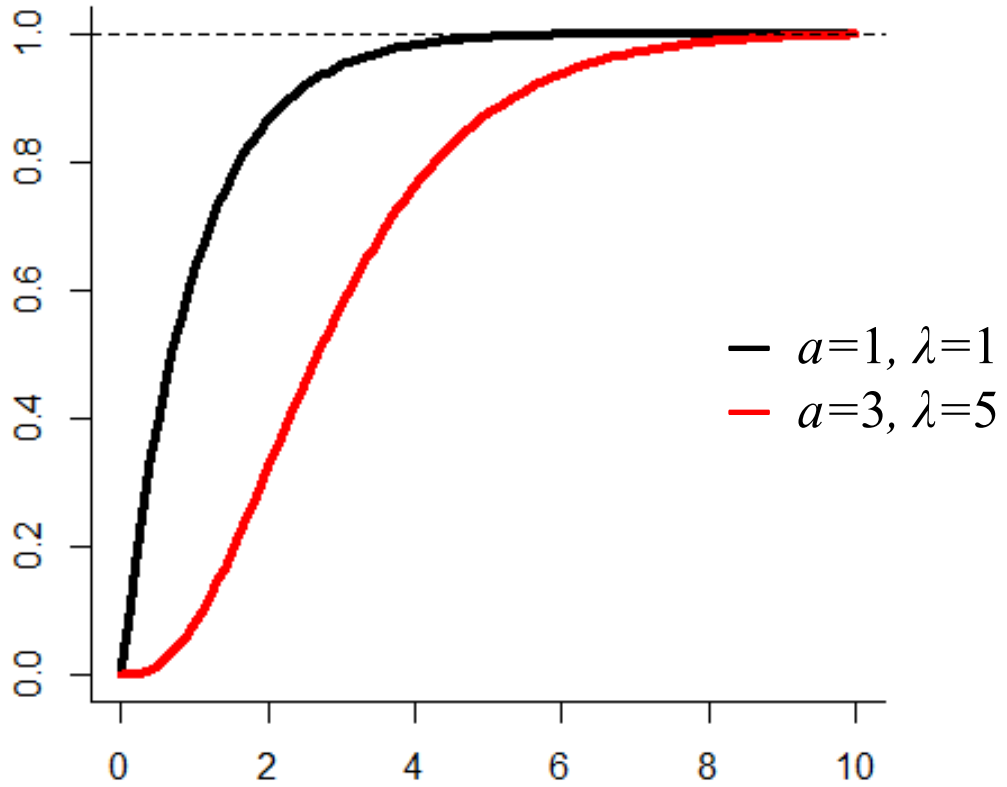
- Defined in terms of *gamma function* $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x}$
  - For $a > 1 \rightarrow \Gamma(a) = (a-1)\Gamma(a-1)$
  - For integer $a \rightarrow \Gamma(a) = (a-1)!$
- No closed-form CDF
  - special case $a = 1 \leftrightarrow$ Exponential(λ)

Gamma Distribution

PDF



CDF



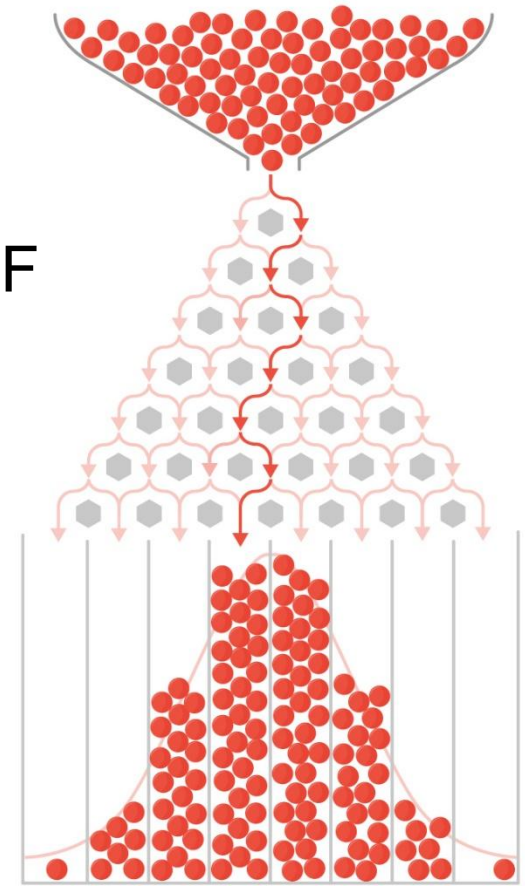
Example

- Let $X \sim$ Poisson(λ) counts #events, and $Y \sim$ Gamma($2,\lambda$) be interval till 2nd event. Show that $\mathrm{P}(X \leq 1) = \mathrm{P}(Y \geq 1)$

Normal Distribution

- *Normal* (a.k.a. *Gaussian*) distribution arises as *limiting distribution* of the average of many independent RVs
  - E.g. as $n \to \infty$, Binomial PMF converges to Normal PDF (De Moivre–Laplace theorem)

- Basic model for
  - Natural phenomena
  - Measurement errors
  - Statistics



Normal Distribution

- RV X follows *Normal distribution* with parameters mean μ and variance σ^2 , denoted by $X \sim N(\mu, \sigma^2)$, if it has the following PDF

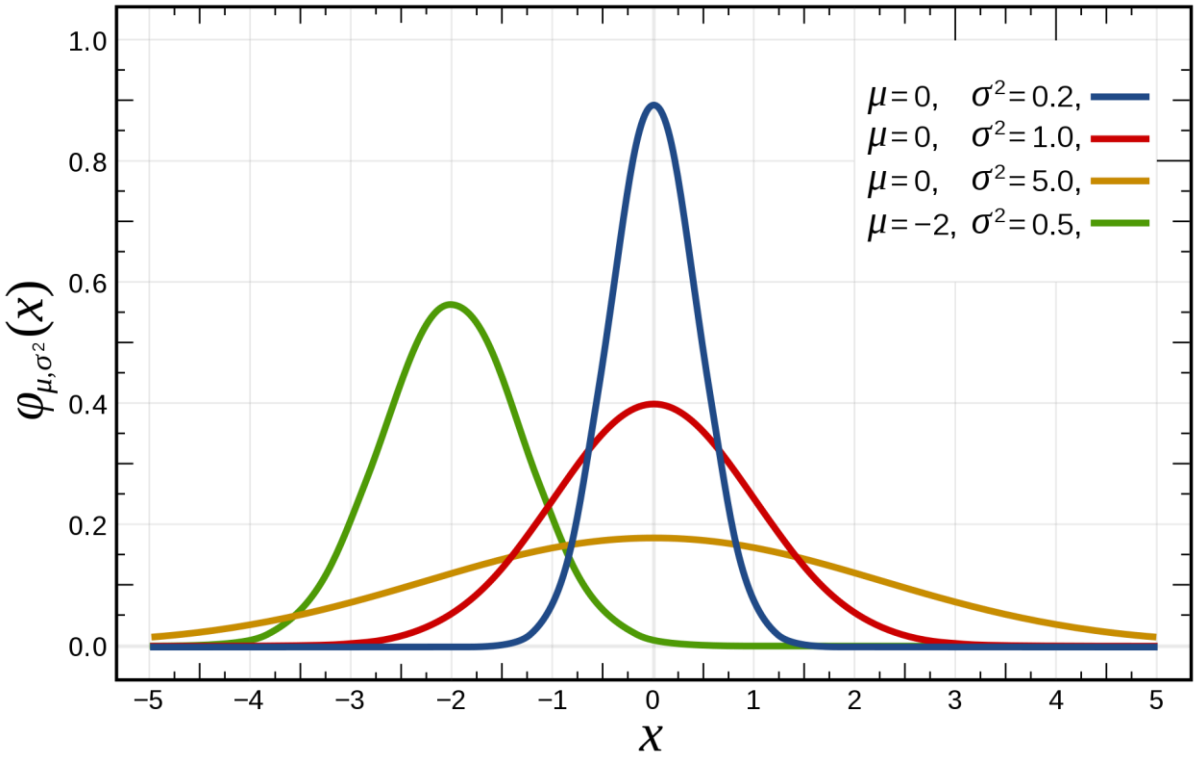
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}, x \in \mathbb{R}$$

  - Parameters $\begin{cases} \mu \in \mathbb{R} & \text{called } \textit{mean}\text{, represents center} \\ \sigma > 0 & \text{called } \textit{standard deviation}\text{, represents spread} \end{cases}$
  - Special case Normal($\mu=0, \sigma=1$) called *standard Normal*

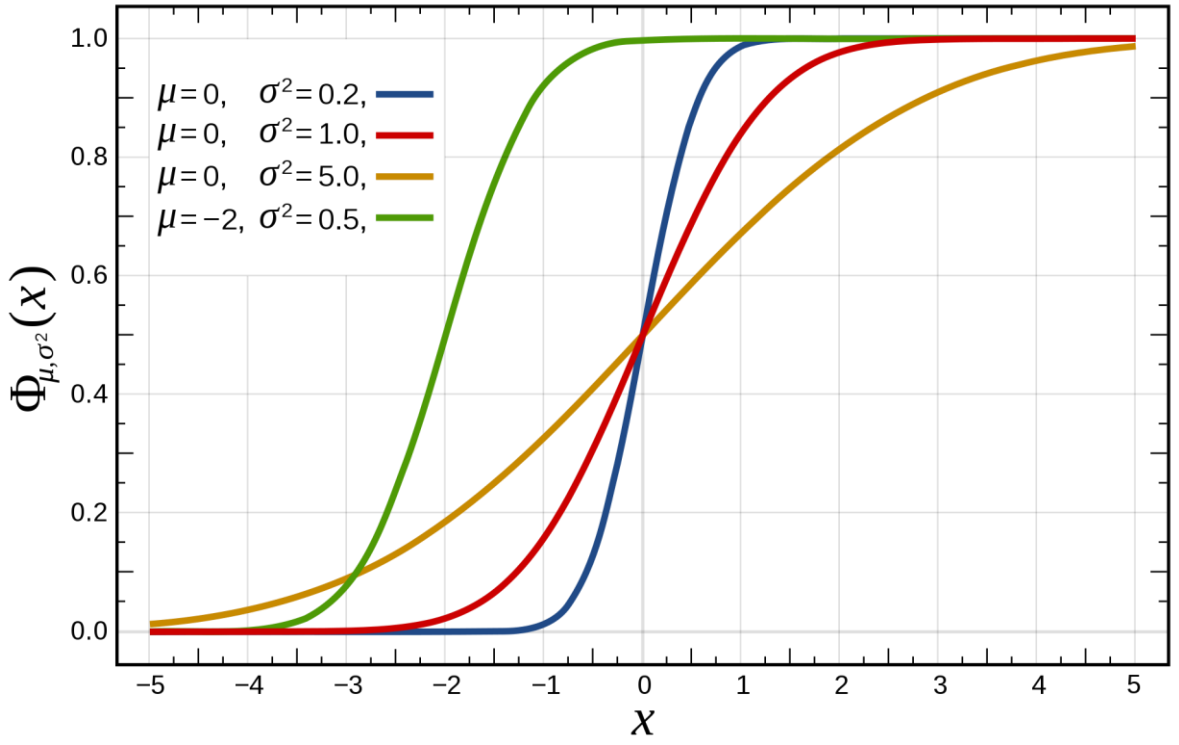
- No closed-form CDF, calculate probabilities using software

Normal Distribution

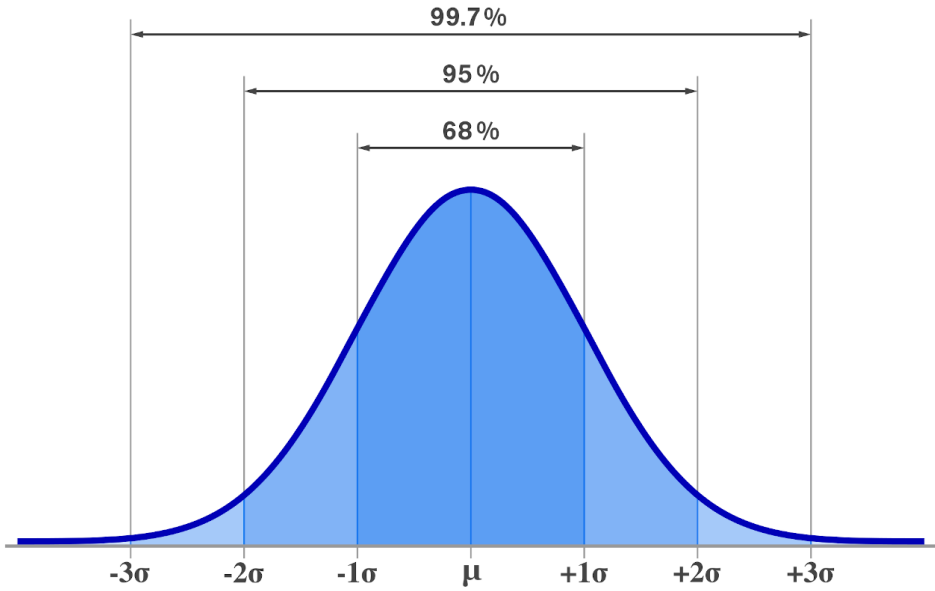
PDF



CDF



Normal Distribution



- 68-95-99 mnemonic rule

$$P(\mu - 1\sigma \le X \le \mu + 1\sigma) = 68.27\%$$

$$P(\mu - 2\sigma \le X \le \mu + 2\sigma) = 95.45\%$$

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) = 99.73\%$$

  - E.g Mensa IQ test has μ=100 & σ =15. If you scored 130, which percentile are you on?